

Trainings ▾

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

WARNING

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

CAUTION

Use caution when draining oil or replacing filters that the oil is not spilled into the bilge area. The oil and filters must be disposed of in accordance with local environmental regulations.

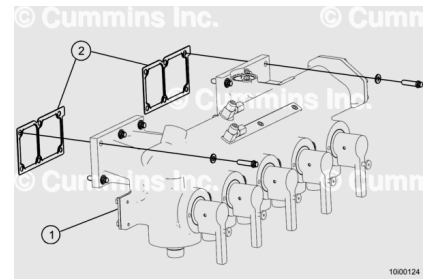
- Disconnect the batteries. Refer to Procedure 013-009 in Section 13.
(/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Remove the lubricating oil filters. Refer to Procedure 007-013 in Section 7.
(/qs3/pubsys2/xml/en/procedures/28/28-007-013-tr.html)

Remove

with Mechanically Actuated Injector

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



The illustration shown is the lubricating oil filter head assembly installed on the left bank.

Note : Only K38 engines with aluminum filter heads have the two studs (1) and nuts.

The use of guide bolts will aid the disassembly procedure.

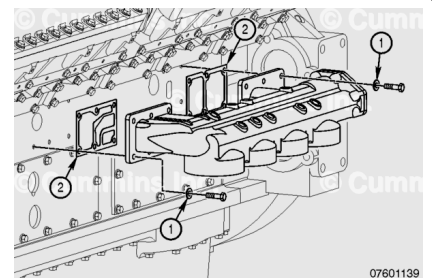
Remove the 12 or 10 capscrews and two nuts.

Remove the oil filter head (1). Remove and discard the gaskets (2).

with Electronically Actuated Injector

⚠ WARNING ⚠

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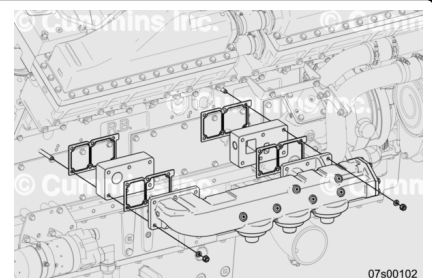


Remove the 12 capscrews and washers (1) securing the filter head to the engine block.

Remove the oil filter head assembly and discard the gaskets (2).

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



If equipped with oil filter head and pre-lubricating oil pump both on right bank, perform the following actions:

- Remove 12 nuts and washers from the ends of the metal studs securing the filter head to the block.
- Remove the oil filter heads assembly and discard the gaskets.

- Remove both left and right filter head adapters. Inspect for reuse.
- Discard of gaskets between adapters and block.
- Remove 12 threaded studs from the block. Inspect for reuse.

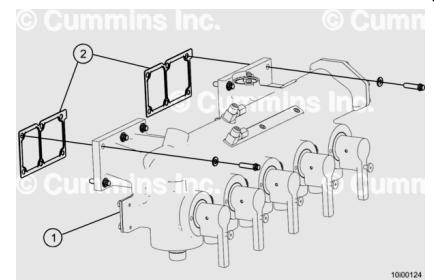
With Duplex Lubricating Oil Filter Head

If necessary, disconnect the hoses to the bypass oil filter.

The use of guide bolts will aid the disassembly procedure.

Remove the 12 capscrews.

Remove the oil filter head (1). Remove and discard the gaskets (2).

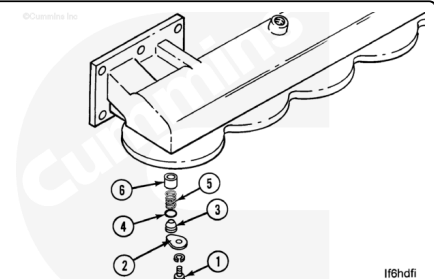


Disassemble

with Mechanically Actuated Injector

⚠ WARNING ⚠

The parts are under spring pressure. The parts can move suddenly with enough force to cause personal injury. Use a tool that will control the parts firmly.



Note : The K50 cast iron oil filter head is shown as an example. The procedure for the K38 cast iron filter head is the same. The filter head housing is the only part that is different for K38 and K50 engines.

Note : Newer filter heads have a straight thread plug and copper washer. Older filter heads can have a heavy plain washer or a retainer tab. The newer filter head does **not** have the stop and o-ring.

Note : Engines with inboard piston cooling nozzles do **not** require the piston cooling regulator.

Remove the:

1. Capscrew and lock washer
2. Retainer
3. Spring stop
4. O-ring
5. Piston cooling regulator valve spring

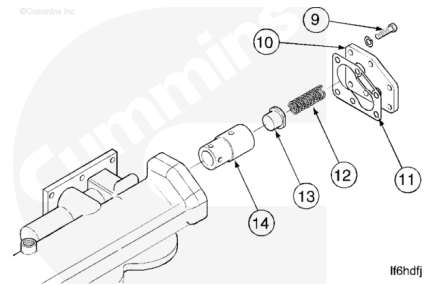
6. Plunger.

Discard the o-ring.

Keep the parts together to aid future installation.

⚠ WARNING ⚠

The parts are under spring pressure. The parts can move suddenly with enough force to cause personal injury. Use a tool that will control the parts firmly.



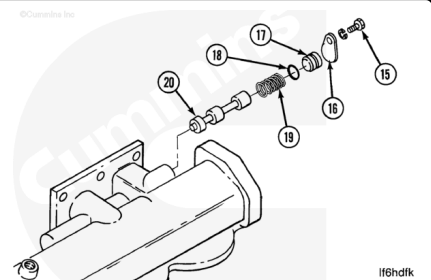
Remove the:

- 9. Capscrews and lock washers (five each)
- 10. Cover plate
- 11. Gasket
- 12. Filter bypass valve spring
- 13. Plunger
- 14. Housing.

Keep the parts together to aid future installation.

⚠ WARNING ⚠

The parts are under spring pressure. The parts can move suddenly with enough force to cause personal injury. Use a tool that will control the parts firmly.



Note : Newer filter heads have a straight thread plug and a copper washer. Older filter heads can have a heavy plain washer or a retainer tab. The newer filter head does not have the spring stop and o-ring.

Remove the:

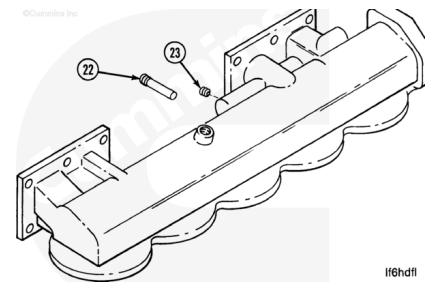
- 15. Capscrew and lock washer
- 16. Retainer
- 17. Spring stop
- 18. O-ring
- 19. Pressure regulator valve spring
- 20. Plunger.

Discard the o-ring.

Note : If the plug is a restriction-type plug (22), it must be installed again. The K38 filter head does not contain

the restriction plug.

Remove the pipe plug (23) or the restriction plug.



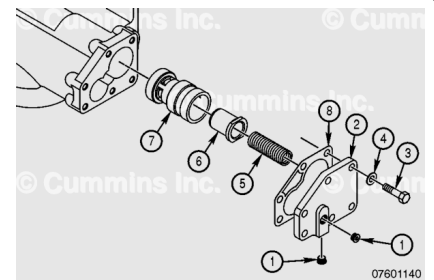
with Electronically Actuated Injector

Remove both pipe plugs (1).

Remove the filter bypass valve cover plate (2) by removing all six capscrews and washers (3, 4).

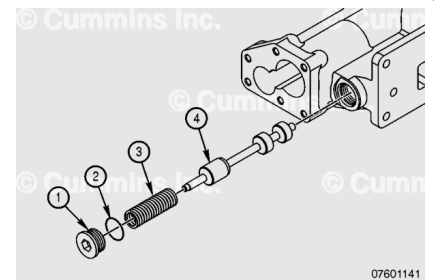
Discard the gasket (8).

Remove the compression spring (5), plunger (6), and bypass valve body (7).



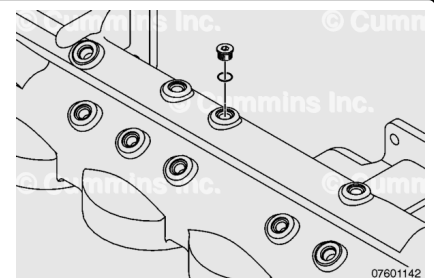
Remove the threaded plug and discard the o-ring (1, 2).

Remove the compression spring (3), and pressure regulator plunger (4).



Remove the remaining straight thread o-ring plugs.

Remove any straight thread o-ring fittings if used.



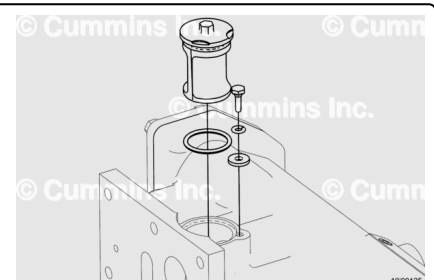
With Duplex Lubricating Oil Filter Head

Note the location of the machined notches on top of the shut off valve spool.

Remove the capscrew, lock washer and flat washer.

Remove the valve spool.

Remove and discard the o-ring.

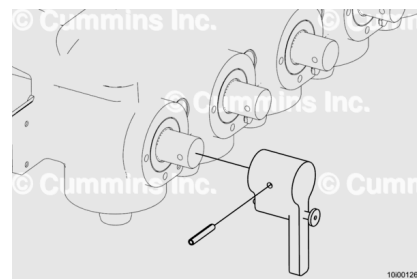


Use a pin punch of the appropriate size and a hammer.

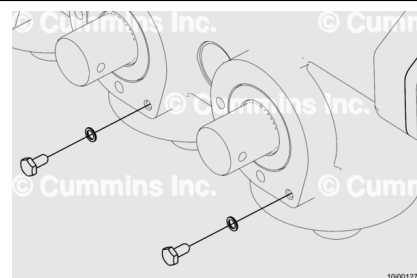
Drive the roll pin out of the five filter head levers.

Remove the levers from the plungers.

The detent rod, spring and knurled knob do **not** need to be removed from the lever unless the spring is broken, or the rod or knob are damaged.



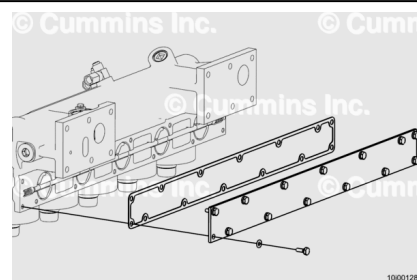
Remove the five bleed screws and washers.



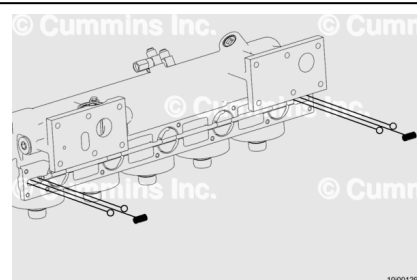
Position the filter head assembly so that the cover plate parallel to the floor. There are loose parts behind the cover that can fall out.

Remove the capscrews that attach the cover head to the filter head.

Remove and discard the gasket.

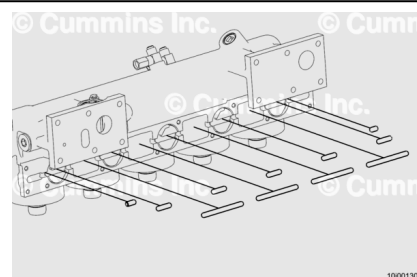


Carefully remove the four bearing balls and two springs from the filter head.



Note the length and orientation of the links.

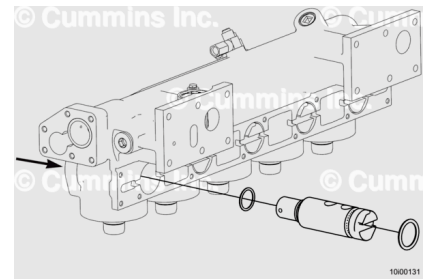
Remove the 10 links.



Remove all five plungers.

Remove and discard the o-rings.

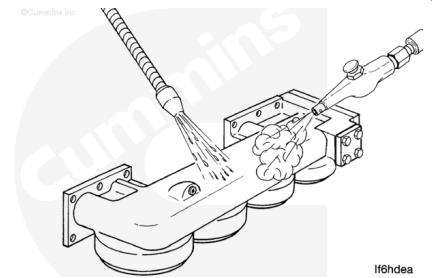
Remove all the straight thread o-ring plugs. Discard the o-rings.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the risk of personal injury.



⚠ WARNING ⚠

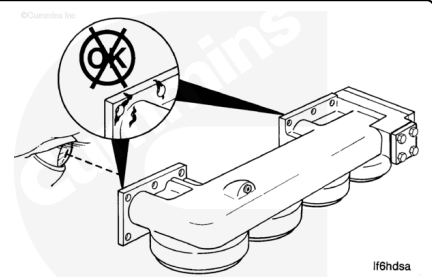
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

If the gaskets are stuck to the filter head assembly, use a gasket scraper. Remove the two mounting gaskets.

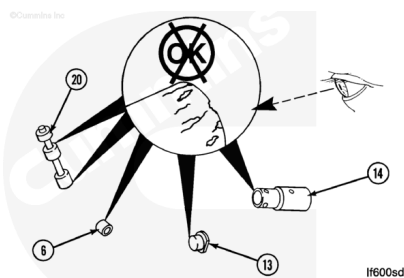
Use solvent to clean the lubricating oil filter head assembly.

Dry with compressed air.

Check the lubricating oil filter head for cracks or other damage. Check the visible portion of the valve plungers for signs of scoring or other damage.



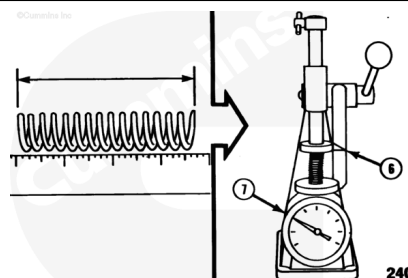
Check the surfaces of the pressure regulator and valve plungers and their mating bores in the housings or the filter head. The part **must** be replaced if it is **not** possible to remove the scratches or marks with a fine crocus cloth.



Measure the free length of the spring. Use a spring tester.
Measure the force of the spring (7) at the working height (6).

Filter Bypass Valve Spring (12), and Pressure
Regulator valve Spring (19) Part Number 205659

	mm		in
Free Length	102	MIN	4.03
	106	MAX	4.18
Working Height (6)	63.5	NOM	2.50
Load (7)	249	MIN	56.0
	286	MAX	64.4



Piston Cooling Regulator Valve Spring (5), Part
Number 205349

	mm		in
Free Length	60.7	MIN	2.39
	64.5	MAX	2.54
Working Height (6)	44.4	NOM	1.75
Load (7)	51	MIN	11.5
	57	MAX	12.8

If any of the springs are **not** within specifications, the part **must** be replaced.

Assemble

with Mechanically Actuated Injector

Note : Only a portion of the filter head is illustrated for clarity.

Note : Do not remove the adapters unless they are damaged or loose.

Check the filter adapters (24).

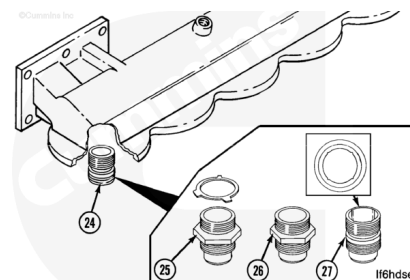
Note : Four types of adapters have been used. The adapter that requires a lock plate (25), and the adapter with the external hex (26), an adapter with internal grooves (27) and an adapter with increased engagement. The internal grooves allow the use of a section of [0.750 in] square keystock to turn and tighten the part.

If the adapter **must** be replaced, Loctite™ 270 or equivalent, **must** be put on the threads that contact the threads in the filter head.

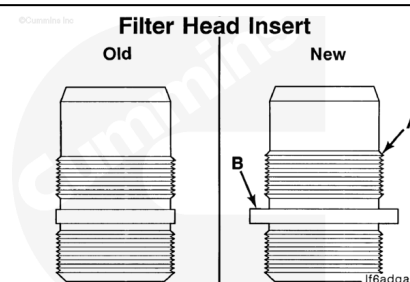
Torque Value: 95 n•m [70 ft-lb]

If the older adapters with the external hex are installed, put Loctite™ 270 on the threads.

Torque Value: 200 n•m [148 ft-lb]

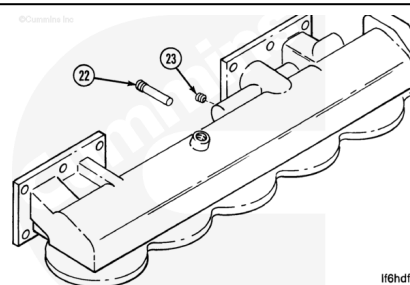


Note : All K38 and K50 engines produced after July, 1990, ESN first 33118127, have an adapter fitted with increased thread engagement. This change was made to confirm that a minimum of three full threads are engaged.



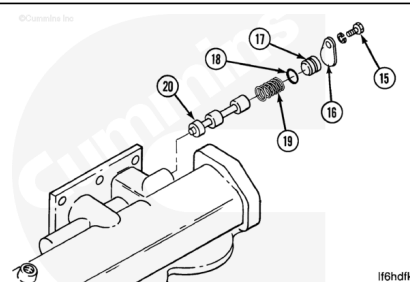
Install the pipe plug (23) or the restriction plug (22).

Torque Value: 20 n•m [15 ft-lb]



⚠ WARNING ⚠

The parts are under spring pressure. The parts can move suddenly with enough force to cause personal injury. Use a tool that will control the parts firmly.



Note : If a straight thread plug and a copper washer retain the spring, the spring stop and o-ring are not installed.

Use clean engine oil. Lubricate the pressure regulator plunger (20).

Install the plunger in the filter head bore.

Be sure the part moves freely in the housing. If the part does **not** move freely, check for debris and burrs on the part and assemble again.

Install the o-ring (18) on the spring stop (17).

Use clean engine oil. Lubricate the o-ring.

Install the pressure regulator spring (19).

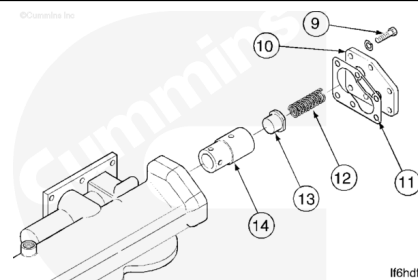
Install the spring stop (17), the retainer (16), and the lock washer and capscrew (15).

Torque Value: 47 n•m [35 ft-lb]

Note : If the straight thread plug and copper washer are installed, tighten the plug to the value indicated above.

⚠ WARNING ⚠

The parts are under spring pressure. The parts can move suddenly with enough force to cause personal injury. Use a tool that will control the parts firmly.



Use clean engine oil. Lubricate the filter bypass valve plunger (13). Install the plunger in the housing (14).

Be sure the plunger moves freely in the housing. If the part does **not** move freely, check for debris or burrs on the part and assemble again.

Install the filter bypass valve housing and plunger in the filter head.

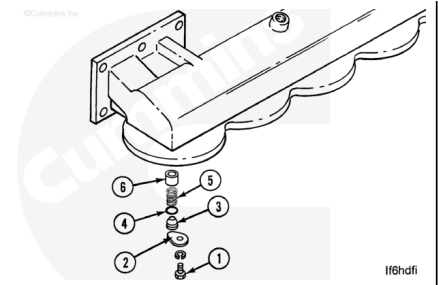
Install the:

- 12. Spring
- 11. Gasket
- 10. Cover plate
- 9. Lock washers and capscrews (five each)

Torque Value: 41 n•m [30 ft-lb]

⚠ WARNING ⚠

The parts are under spring pressure. The parts can move suddenly with enough force to cause personal injury. Use a tool that will control the parts firmly.



Note : If a straight thread plug and copper washer retain the spring, the spring stop and o-ring are not required.

Use clean engine oil. Lubricate the piston cooling regulator plunger (6).

Install the plunger in the bore. Be sure the plunger moves freely. If the plunger does **not** move freely, check for debris or burrs on the part and assemble again.

Install the o-ring (4) on the spring stop (3).

Use clean engine oil. Lubricate the o-ring.

Note : Engines with inboard piston cooling nozzles do not require the piston cooling regulator.

Install the:

- 5. Spring
- 3. Spring stop
- 2. Retainer
- 1. Lock washer and capscrew

Torque Value: 47 n·m [35 ft-lb]

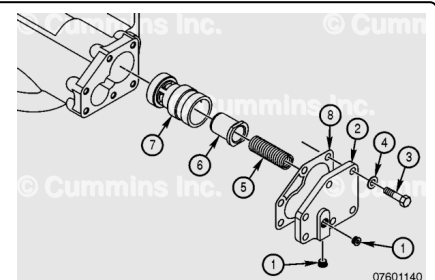
Note : If a straight thread plug and copper washer are installed, tighten the plug to the value indicated above.

with Electronically Actuated Injector

Use clean engine oil to lubricate the bypass valve and related parts.

Install the bypass valve body (7), plunger (6), and compression spring (5).

Install a new gasket (8), and the filter bypass valve cover plate (2) by installing all six capscrews and washers (3, 4).



Torque Value: 47 n·m [35 ft-lb]

Install both pipe plugs (1).

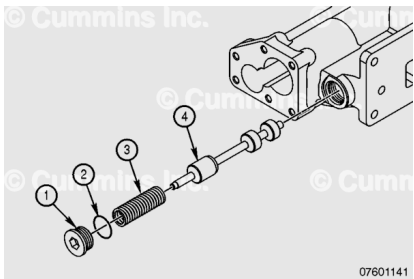
Torque Value: 20 n·m [177 in-lb]

Use clean engine oil to lubricate the regulator plunger and related parts.

Install the pressure regulator plunger (4), and compression spring (3).

Install the threaded plug and new o-ring (1, 2).

Torque Value: 122 n•m [90 ft-lb]



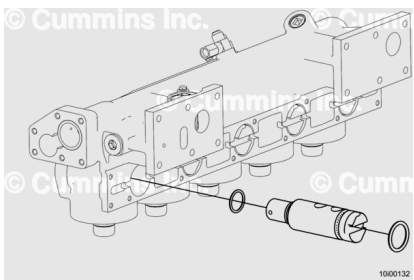
With Duplex Lubricating Oil Filter Head

Install two o-rings on each valve plunger.

Use engine oil.

Lubricate the plunger, the o-rings and the bore in the filter head.

Install all five valve plungers.



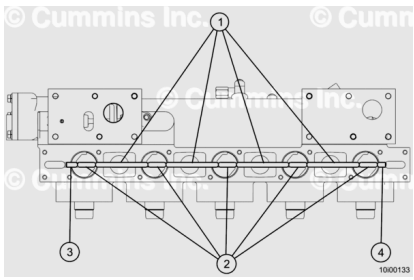
Position the filter head so that the face for the cover is parallel to the floor so the loose parts will remain in position.

Use Lubriplate™ 105 or equivalent to hold the parts in position.

Install the links.

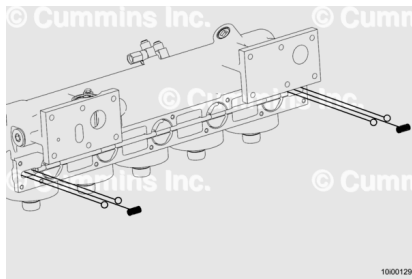
See the table below for detail on position and orientation.

The lengths are given for identification **only**. The lengths have been rounded and are **not** specifications.



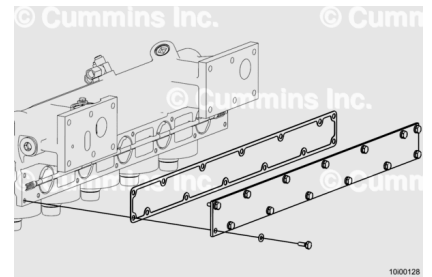
Callout	Length mm [in]	Position
1	103 [4]	Between the plungers.
2	36 [1.4]	On the end of plungers.
3	22 [0.8]	On the left end. Flat end toward ball/spring groove.
4	17 [0.7]	On the right end. Flat end toward the ball/spring groove.

Install the four bearing balls and two springs in the slots in the filter head.



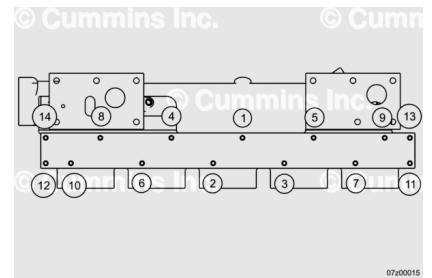
Install the cover gasket and cover.

Install the capscrews and flat washers.



Tighten the capscrews in the sequence shown.

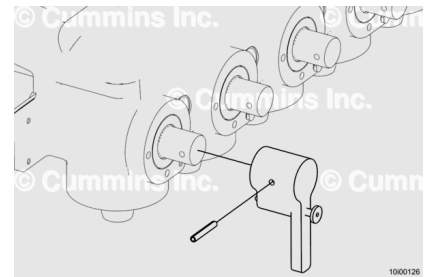
Torque Value: 27 n•m [20 ft-lb]



Verify the detent spring is **not** broken by pressing on the detent rod and look to see if it returns to position protruding from the handle

Install the handles on the plungers.

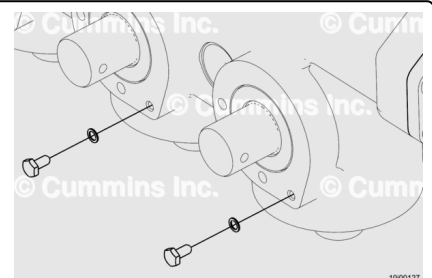
Drive the roll pin through the handle and the hole in the plunger until it is slightly below the surface on both sides of the handle.



Install the five bleed screws and washers.

Tighten the capscrews.

Torque Value: 14 n•m [120 in-lb]



Install the o-ring on the shut-off valve spool.

Lubricate the o-ring and the bore in the filter head with engine oil.

Install the spool in the bore.

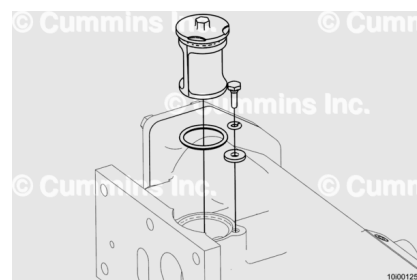
One notch should be close to the body of the filter head, about the 10 o'clock position.

One notch should be aligned with the threaded hole, about the 8 o'clock position.

Install the flat washer, lock washer and capscrew.

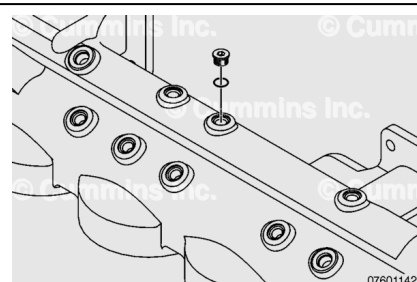
Tighten the capscrew.

Torque Value: 15 n•m [132 in-lb]



Install the remaining straight thread o-ring plugs.

Torque Value: 20 n•m [177 in-lb]

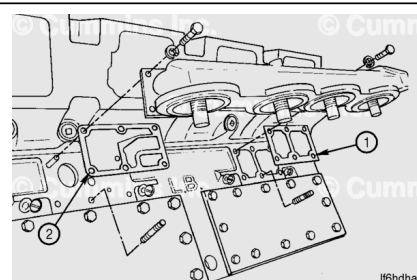


Install

with Mechanically Actuated Injector

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

The oil filter head can be installed on either engine bank. Verify the lubricating oil transfer tube has been installed correctly. The oil filter head must be installed **OPPOSITE** the bank to where the oil transfer tube is connected. If the oil filter is installed on the wrong bank, the engine will be damaged severely because of lack of lubrication.

⚠ CAUTION ⚠

The gaskets must be installed as shown or the engine can be damaged because of lack of lubrication.

In the illustration, the filter head is being installed on the left bank.

Note : Only a K38 engine with an oil filter head manufactured from aluminum requires two studs.

Install the two studs (1).

The use of guide bolts will aid the assembly procedure.

The gasket (3) has a rib down the center. The gasket is installed for the right end **only** of the oil filter head.

Install the gaskets (2) and (3).

Note : Engines with outboard piston cooling nozzles must have two gaskets (2 and 3) installed, as described and illustrated above. Engines with inboard piston cooling nozzles can have the same gasket (3) installed to both locations. Make sure all six mounting holes are visible through each gasket.

Install the oil filter head (4).

Note : Two nuts are required on a K38 engine only with a filter head manufactured from aluminum.

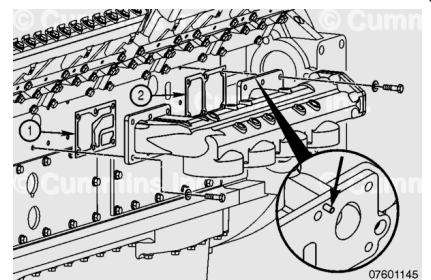
Install the 12 lock washers and capscrews.

Torque Value: 47 n•m [35 ft•lb]

with Electronically Actuated Injector

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

The oil filter head can be installed on either engine bank. Verify the lubricating oil transfer tube has been installed correctly. The oil filter head must be installed OPPOSITE the bank where the oil transfer tube is connected. If the oil filter is installed on the wrong bank, the engine will be damaged severely because of lack of lubrication.

⚠ CAUTION ⚠

The gaskets must be installed as shown or the engine can be damaged because of lack of lubrication.

Note : The QSK50 uses a single dowel pin. Make certain that the dowel pin is engaged in place when the lubricating oil filter head assembly is installed.

Install the two new lubricating oil filter head assembly gaskets (2).

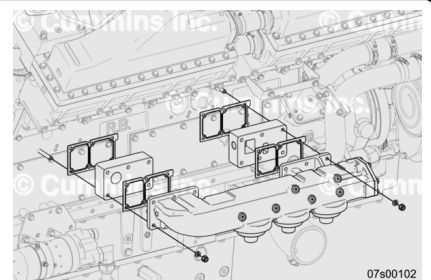
Note : The same gasket (2) can be installed in both locations. Make sure the six mounting holes are visible through each gasket.

Install the lubricating oil filter head assembly and twelve capscrews.

Torque Value: 47 n•m [35 ft-lb]

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

The oil filter head can be installed on either engine bank. Verify the lubricating oil transfer tube has been installed correctly. The oil filter head must be installed **OPPOSITE** the bank where the oil transfer tube is connected. If the oil filter is installed on the wrong bank, the engine will be damaged severely because of lack of lubrication.

If equipped with oil filter head and pre-lubricating oil pump both on right bank, perform the following actions:

Install 12 threaded studs with Loctite® 242 applied to the block.

Tighten using a stud runner.

Torque Value: 53 n•m [39 ft-lb]

Install the two new lubricating oil filter head gaskets against the block.

Install both left and right filter head adapters.

Install the next two lubricating oil filter head gaskets against the filter head adapters.

Install all 12 nuts to the end of the studs and tighten.

Torque Value: 53 n•m [39 ft-lb]

With Duplex Lubricating Oil Filter Head

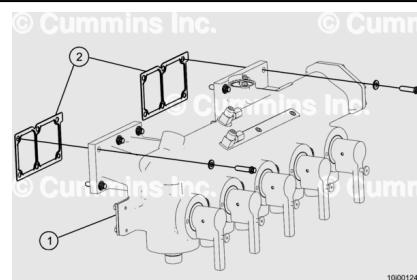
The use of guide bolts will aid the assembly procedure.

Install the gaskets (1) and (2).

Install the oil filter head.

Install the 12 lock washers and capscrews.

Torque Value: 47 n•m [35 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

Use caution when draining oil or replacing filters that the oil is not spilled into the bilge area. The oil and filters must be disposed of in accordance with local environmental regulations.

- Install the lubricating oil filters. Refer to Procedure 007-013 in Section 7.
(/qs3/pubsys2/xml/en/procedures/28/28-007-013-tr.html)
- Connect the bypass filter hoses, if equipped.
- Connect the batteries. Refer to Procedure 013-009 in Section 13.
(/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)

- Operate the engine to normal operating temperature. Check for leaks.

Last Modified: 15-May-2023
